LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray
5. Specific Objectives	19.1.2.1 Trainee will be able to explain principle of radiographic inspection.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of sample radiographs to identify defects and discuss safety protocols
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of radiographic inspection?
(b) Name two sources used for generating radiation in NDT.
(c) Why is safety precaution critical when handling X-ray equipment?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of radiographic inspection • Interaction of radiation with matter • Shadow formation and motion blur • Types of recording media	L & Ds	PPT, IB	15 min
(b)	• X-ray and Gamma ray sources • Applicability of radiography • Types of test specimens	L & Ds	PPT, PDF	20 min
(c)	• Safety precautions during X-ray use • Regulatory limits for exposure • Emergency procedures for radiation leaks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a radiographic shadow indicates a defect in a specimen.
(b) Differentiate between X-ray and Gamma ray sources in terms of applicability.
(c) List three essential safety precautions when operating X-ray equipment.

12. Home Assignment and Follow Up	Prepare a brief report comparing the principle and applications of neutron radiography discussed in the next lesson (LP 06) with the X-ray radiography covered in this session.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Inspection of neutron radiography introduction, different methods principle and various applications	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min